# Rex Tin Long Fung

E-mail: rexfung@umich.edu
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## EDUCATION

**University of Michigan**, Ann Arbor, MI — Sep 2022 - Expected Sep 2027
PhD Candidate specializing in MRI sequence programming and iterative reconstruction methods, with applications to accelerating functional MRI.

**University of Michigan**, Ann Arbor, MI — Sep 2022 - Aug 2024
**Program**: MS, Biomedical Engineering (Imaging and Ultrasonics) **GPA**: 3.65/4.00
**Relevant Coursework**: Linear Algebra, Optimization, Probability, Machine Learning, Image Processing, Medical Imaging Systems

**University of Michigan**, Ann Arbor, MI — Sep 2018 - May 2022
**Program**: BS, Neuroscience major, Electrical Engineering minor **GPA**: 3.78/4.00
**Relevant Coursework**: Digital Signal Processing, Electrical Biophysics, Embedded Control Systems, Sensing and ML for Neural Interfaces

## RESEARCH EXPERIENCE

**Functional MRI Lab**, University of Michigan — Sep 2022 - Present
Graduate Student Research Assistant — *Advisors: Dr. Jon-Fredrik Nielsen, Dr. Douglas Noll*

- Accelerated fMRI data acquisition by 10× by developing a novel 3D-EPI sequence with Poisson-disc random sampling, increasing resolution for functional neuroimaging studies.
- Simultaneously reconstructed 100 GB+ fMRI datasets on HPCs using a custom 4D iterative optimization algorithm written in Julia and Python, which models low-rank signal structures on multiple scales to improve accuracy of functional activation maps.
- Standardized cross-vendor research for 7+ international institutions (e.g., Harvard, Freiburg) by developing open-source Pulseq sequences, facilitating seamless, hardware-agnostic protocol sharing.
- Authored 1 first-author manuscript (under revision) and co-authored 1 journal article and 4 conference presentations at top-tier venues (e.g. IEEE, ISMRM), translating complex sequence design and reconstruction findings for the global imaging community.

**Bo Duan Lab**, University of Michigan — Jul 2020 - May 2022
Undergraduate Research Assistant — *Advisor: Dr. Bo Duan*

- Reduced manual data processing time by 90% by using OpenCV to automate behavioral tracking in 30+ hours of footage and cell-counting for 200+ microscopy images.
- Co-authored 1 journal article in *Nature Communications*, characterizing a dedicated skin-to-brain neural circuit for cool sensation in mice.

## TECHNICAL SKILLS

**Math & Engineering**: Signal Processing, Inverse Problems, Optimization, Machine Learning
**Programming**: Julia, Python, MATLAB, GPU programming, Jupyter Notebooks, LaTeX, Git/GitHub
**Science**: Experimental Design, Academic Writing and Presentations

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## JOURNAL PUBLICATIONS

[1] R.A. Lobos, X. Wang, **R.T.L. Fung**, Y. He, D. Frey, D. Gupta, Z. Liu, J.A. Fessler, D.C. Noll. "Spatiotemporal Maps for Dynamic MRI Reconstruction.", *IEEE Transactions on Computational Imaging* (2026). DOI.

[2] H. Lee, C.C. Hor, L.R. Horwitz, A. Xiong, X.Y. Su, D.R. Soden, S. Yang, W. Cai, W. Zhang, C. Li, C. Radcliff, A. Dinh, **T.L.R. Fung**, I. Rovcanin, K.P. Pipe, X.Z.S. Xu, B. Duan. "A dedicated skin-to-brain circuit for cool sensation in mice.", *Nature Communications* (2025). DOI.

## MANUSCRIPTS UNDER REVISION

[R1] **R.T.L. Fung**, R.A. Lobos, J.A. Fessler, D.C. Noll, J.-F. Nielsen. "ArbEPI: 3D-EPI sequences from arbitrary 2D sampling masks, with multiscale low-rank reconstruction for fMRI.", *Magnetic Resonance in Medicine*, under revision.

## CONFERENCE POSTERS

[1] **R.T.L. Fung**, D.C. Noll, J.-F. Nielsen. "ArbEPI: A Framework for Generating 3D-EPI Sequences from Arbitrary Sampling Patterns" *In Vivo Magnetic Resonance, Gordon Research Conferences*, 2026. 20 minute talk and traditional poster.

[2] **R.T.L. Fung**, R.A. Lobos, J.A. Fessler, D.C. Noll, J.-F. Nielsen. "Sub 2 mm Resolution fMRI at 3T using Randomly Undersampled 3D-EPI with Locally Low-Rank + Temporally Sparse Reconstruction" *International Society for Magnetic Resonance in Medicine Annual Meeting*, 2025. Digital poster #3196.

[3] S. Peltier, M. Egan, **R. Fung**, Q. Chen, M. Zaitsev, J.-F. Nielsen. "Harmonization of an SMS-EPI fMRI protocol using Pulseq: Sequence implementation and ABCD QC metrics." *Organization for Human Brain Mapping Annual Meeting*, 2024. Digital poster #3804.

[4] Y. He, **R. Fung**, J.-F. Nielsen. "High-Accuracy Ultra-short Inner-Volume Saturation Pulse for 3D Steady-State Imaging." *International Society for Magnetic Resonance in Medicine Annual Meeting*, 2024. Digital poster #3249.

## TEACHING EXPERIENCE

**BME/ECE 516 Medical Imaging Systems**, University of Michigan Sep 2024 - Dec 2024

- Mentored 20-30 students to learn Medical Imaging Systems e.g. MRI, ultrasound, by guiding them through homework problems during weekly office hours and via Piazza, an online Q&A forum.
- Streamlined course logistics for 15 weeks by releasing and grading homework assignments + final exam on Canvas, an online site for the class.

## HONORS AND AWARDS

**James B. Angell Scholar**, University of Michigan Mar 2021
*Students who achieve an "A" record (all grades of A+, A, or A-) for two or more consecutive terms*

## OTHER SKILLS

**Digital Media Production**: Adobe Premiere Pro/Rush, Audacity, Edits by Instagram
**Languages**: English, Cantonese, Mandarin Chinese, Japanese
**Sports**: Volleyball, Bouldering, Badminton, Pickleball, Table Tennis, Basketball, Triathlon